



No one-size-fits-all: a study of prompt techniques and large language models to enhance AI's mathematics educational quality

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Abstract

Large Language Models (LLMs) are increasingly discussed as tools for supporting mathematical problem solving. However, existing research predominantly evaluates LLM performance in terms of correctness, while the mathematics educational quality of AI-generated worked-out solutions remains largely unexplored. This study investigates how the mathematics educational quality (a set of quality criteria relating to teaching and learning mathematics) of AI-generated problem solutions can be influenced by model choice and prompt engineering. Four LLMs (Gemini 1.5 Pro (Advanced), Claude 3.5 Sonnet, ChatGPT-o3 mini, and DeepSeek-R1) were tested on six problem-solving tasks from the domains of number and algebra using four prompt techniques (Zero Shot, Chain of Thought, Persona, and Retrieval-Augmented Generation). In total, 2880 solutions were analyzed, combining human expert coding of content-related, process-related, and pedagogical-contextual quality with binary logistic regression. Results show that content-related quality is mainly driven by model type and task characteristics, whereas process-related and pedagogical-contextual quality depend strongly on prompt design, particularly Persona prompting. Overall, no single model or prompt technique performs optimally across all dimensions, indicating that effective educational use of LLMs requires context-sensitive combinations of models, prompts, and tasks.

Keywords Large language models · Generative artificial intelligence · Problem solving · Prompt engineering · Number · Algebra

1 Introduction

Artificial intelligence technology has long been used in mathematics education, particularly through learning analytics and intelligent tutoring systems (Shin, 2020). More recently, generative AI based on large language models (LLMs) has considerably expanded the scope of AI applications in this field (Pepin et al., 2025; Richard et al., 2022). In mathematical problem solving, LLMs can generate worked examples, provide feedback, and potentially support students' reasoning processes (Jia et al., 2024; Park & Manley, 2024; Song et al., 2024).

Despite this potential, their educational use poses notable challenges. A central concern is hallucination, that is, the

generation of incorrect or invented information due to biased or insufficient training data (Rawte et al., 2023; Zhang et al., 2024). Although recent reasoning models, which are instructed by internal system prompts to decompose the users' input into intermediate steps, attempt to mitigate this, hallucinations still make careful review of AI outputs by human experts essential. Especially in mathematics, hallucinations can lead to severe errors that cascade through subsequent calculation steps (Chang et al., 2024; Lightman et al., 2023; Plevris et al., 2023). In addition, systematic training-data bias may steer models toward frequent or conventional solution paths to mathematics tasks (Ali et al., 2024), alongside broader AI risks such as misleading objectives or concentration of power (Hendrycks et al., 2023).

For the use of AI systems in mathematics classrooms, it therefore remains essential not only to assess whether LLMs produce correct answers. Equally important is *how* they present their solutions, as solution pathways reveal whether models support learners in developing strategic reasoning and conceptual insight, rather than merely arriving at an answer. Comparative evaluations of different LLMs'

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problem-solving performance are therefore critical for their ethically responsible use in mathematics education. Existing benchmarks primarily assess correctness and logical consistency of the reasoning (Cobbe et al., 2021; Hendrycks et al., 2021), while studies indicate that LLMs struggle especially with tasks involving numerous or diverse operations, conversions that depend on real-world knowledge, or word problems with redundant or unfeasible quantities and quantitative comparisons (Albornoz-De Luise et al., 2024; Srivatsa & Kochmar, 2024). While the content-related quality of solutions, such as correctness or accuracy, varies across models, process-related qualities, such as strategy use or representational changes, hinge on prompt design and often lack a robust mathematics educational quality (Schorcht et al., 2024). Mathematics educational quality here refers to a set of quality criteria relating to the teaching and learning of mathematics, encompassing the correctness and clarity of solutions (content-related), the use of strategies and representations (process-related), and pedagogical features such as addressing common misconceptions or offering alternative approaches (pedagogical-contextual). While benchmark studies in mathematical problem solving are valuable (e.g., Albornoz-De Luise et al., 2024; Frieder et al., 2023; Srivatsa & Kochmar, 2024; Yang et al., 2023), using LLMs in mathematics educational contexts requires systematically evaluating the outputs based on established mathematics educational quality criteria. Accordingly, there is a notable lack of systematic studies examining the educational quality of AI-generated problem solutions and the factors that may influence it.

This study addresses this gap by investigating how mathematics educational quality in AI-generated worked-out solutions can be enhanced through systematic variation of LLMs and prompt techniques. Due to the dependence of the LLMs' performance on the underlying task, we focus on two mathematical domains and systematically compare earlier LLM versions (standard models) with the latest LLMs (reasoning models) based on 2880 generated solutions. We further investigate whether recent customization approaches improve the quality of solution pathways.

2 Theoretical background

2.1 Problem solving in mathematics education

Since the seminal works of Pólya (1957) and Schoenfeld (1985), mathematical problem solving has been identified as a central activity in mathematics learning. Problem solving arises when students encounter barriers that demand cognitive effort to overcome (Pólya, 1957; Schoenfeld 1985, 1992). Recent overviews on mathematical problem solving emphasize the importance of strategy exploration and heuristics for overcoming these barriers, such as working

backwards, changing representations, or drawing analogies (Hembree, 1992; Liljedahl & Cai, 2021; Törner et al., 2007). Pólya (1957, p. vii) described heuristics as methods of problem solving that, unlike algorithms, do not guarantee solutions but offer productive pathways that should be taught, practiced, and reflected upon. Their successful use depends on metacognitive abilities, such as self-regulation, monitoring, and reflection (Artzt & Armour-Thomas, 1992; Pólya, 1957; Schoenfeld, 1992). There is broad consensus that students must actively engage with problems to develop meaningful understanding (Jacinto & Carreira, 2017; Pólya, 1957; Rott et al., 2021; Schoenfeld, 1985), and that such engagement should be deliberately supported by teachers.

In recent decades, the promotion of problem-solving competence has increasingly relied on mental development and self-regulation, self-explanation prompts, and scaffolded instructional designs (Goos & Galbraith, 1996; Herold-Blasius, 2024; Russo et al., 2020). Worked examples, fully worked-out solutions to representative problems have been shown to reduce cognitive load (Sweller, 1988), support schema acquisition (Sweller & Cooper, 1985), and improve conceptual and procedural understanding (Barbieri et al., 2021; Schwonke et al., 2009). When combined with self-explanation prompts, they further encourage learners to reflect on key solution steps and develop structural understanding (Berthold et al., 2009; Renkl et al., 1998). Against this backdrop, our study explores how LLMs can generate high-quality worked-out solutions that embody core heuristics and make strategic reasoning visible and learnable for students. To examine this systematically, we focus on specific mathematical domains that offer both rich problem-solving opportunities and computational accessibility for LLMs – number and algebra (Yang et al., 2023; Plevris et al., 2023).

2.2 Problems in the domains of number and algebra

The focus on number and algebra is motivated by both pedagogical and methodological considerations. International curriculum frameworks identify these domains as foundational for mathematical literacy and curriculum development (Standing Conference of the Ministers of Education and Cultural Affairs, 2022; National Council of Teachers of Mathematics, 2000), a view reflected by large-scale assessments such as the Programme for International Student Assessment (PISA; Organisation for Economic Co-operation and Development [OECD], 2023) and the Trends in International Mathematics and Science Study (TIMSS; Mullis & Martin, 2017). In particular, PISA emphasizes understanding quantitative situations and algebraic relationships as central components of mathematical literacy (OECD, 2023).

From a problem-solving perspective, both domains offer rich opportunities for strategic diversity and representa-

tional flexibility. Number problems typically involve mental calculation, written algorithms, estimation, or visualization (Hiebert & Carpenter, 1992; Kilpatrick et al., 2001). Algebraic problems require symbolic manipulation, recognizing patterns and functional relationships, generalization, and transitions between different solution approaches, e.g., equation-solving, graphical reasoning, or logical inference (Barbieri et al., 2021; Kieran, 2004). Methodologically, number and algebra are particularly suitable for analyzing LLM-generated solutions. Unlike domains such as geometry or data handling, which often require visual reasoning that remains challenging for LLMs (Kazemi et al., 2023; Trinh et al., 2024), number and algebra tasks typically follow well-defined symbolic and procedural structures (e.g., arithmetic procedures, equation solving algorithms) that current models can process more reliably (Kazemi et al., 2023; Trinh et al., 2024; Wardat et al., 2023).

Based on these considerations, six mathematical problem-solving tasks were selected from established literature: three from the number domain and three from the algebra domain (Cooper & Sweller, 1987; Leonardo de Pisa, 1202; Pólya, 1957; Salkind, 1961; Schoenfeld, 1979; Stacey & MacGregor, 1999). Selection criteria included theoretical grounding in mathematics (education) literature, representativeness of early arithmetic and algebra solution processes, strategic diversity, and a variety of applied heuristics. The number tasks involve set partitioning with indeterminate solutions (Marble Problem), multi-step backward reasoning with iterative calculations (Orchard Problem), and systematic trial-and-error strategies (Pails Problem). The algebra tasks address real-world function modeling (Car Problem), formal proof with modular arithmetic (Division Problem), and equation translation and symbolic manipulation (Unknown Number Problem) (see Appendix 1). The problems span a range of cognitive and heuristic demands, from routine procedures to complex heuristic reasoning, from concrete numerical contexts to abstract symbolic manipulation, and from single-step calculations to multi-stage problem solving.

2.3 Generative AI and problem solving

Building on research in mathematical problem-solving, this section examines how advances in LLMs might affect the generation and pedagogical use of worked-out solutions in mathematics education. With the rapid development of generative AI, LLMs have attracted increasing attention as potential tools for supporting mathematical problem solving (Dao & Le, 2023; Federiakin et al., 2024; Hendrycks et al., 2021; Jia et al., 2023; Lewkowycz et al., 2022; Pepin et al., 2025). Despite notable progress, however, LLMs still struggle to produce solutions that are both consistently mathematically correct and pedagogically meaningful (Hendrycks et al., 2021; Plevris et al., 2023; Rawte et al., 2023). Beyond correctness, high-quality mathematical problem solving also

involves engaging with problem structures, applying heuristics, and making sense of representations (Hiebert & Carpenter, 1992). Whether generative AI can meaningfully support such strategic and reflective reasoning remains an open question. A crucial distinction in this context is between *standard* and *reasoning* models. While standard LLMs (e.g., Claude 3.5 Sonnet, Gemini 1.5 Pro) generate outputs via single-pass inference, mapping inputs directly to responses, which is efficient for content creation, translation, data classification and handling frequently asked questions, reasoning models (e.g., ChatGPT-o3 mini, DeepSeek-R1) explicitly decompose problems into intermediate steps and may incorporate self-verification mechanisms (Wei et al., 2023), helpful for complex tasks, analytical thinking, detailed explanations and decision-making. These architectural differences suggest qualitative variation in accuracy, transparency and mathematical reasoning, and motivate direct comparisons between model types in educational contexts. Additionally, recent studies show that despite their accuracy benefits, reasoning models come with high energy costs (Wilhelm et al., 2025), so that for reasons of sustainability, it may make sense to use more energy-efficient models in problem-solving contexts, given a comparable accuracy.

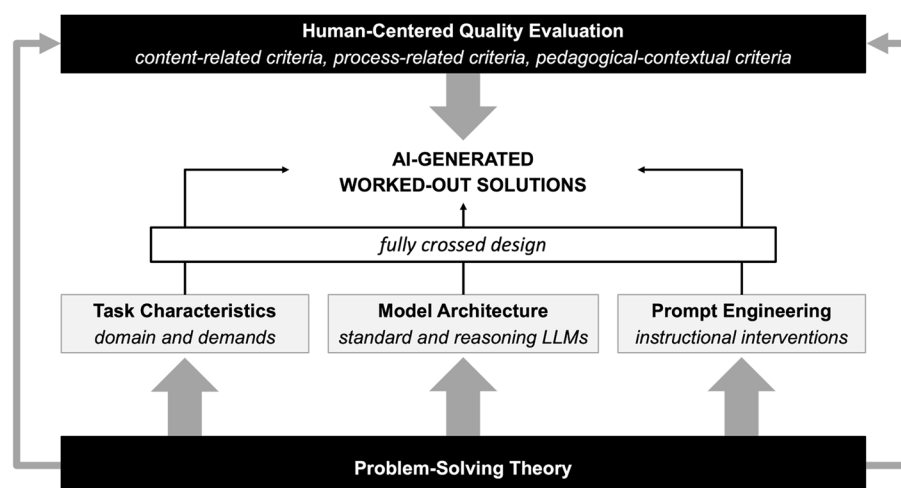
Methodologically, evaluating such differences requires human expert judgment. *Human rating* is therefore a central approach for assessing the appropriateness, accuracy, and pedagogical value of AI-generated responses (Küchemann et al., 2023; Rodriguez-Torrealba et al., 2022; Spreitzer et al., 2024). Model validation studies typically involve systematically entering a series of controlled prompts into LLMs to analyze consistency and sensitivity of model outputs (Tam et al., 2024), which are subsequently assessed by human raters using pre-defined, task-specific criteria (Jacobsen & Weber, 2025; Laskar et al., 2023; Maroengsit et al., 2019; Qiu et al., 2017; Schorcht & Buchholtz, 2024; Schorcht et al., 2024; Spreitzer et al., 2024; Tam et al., 2024). By generating multiple solutions for identical prompts and comparing their rated qualities, the present study examines whether and how LLMs can produce mathematically robust and didactically valuable problem solutions.

2.4 Prompt engineering techniques

Prompt engineering constitutes a central design variable in this study, as prompt formulation shapes not only the correctness, but also the structure and pedagogical orientation of AI-generated mathematical solutions (Schorcht et al., 2024; Vatsal & Dubey, 2024). In mathematical problem solving, prompts can be understood as instructional interventions that guide how LLMs frame, explain, and justify solution pathways.

Four prompt techniques were selected that represent distinct instructional logics relevant to mathematics education.

Fig. 1 Human-centered evaluation framework for educational value assessment of AI-generated worked-out solutions



Zero Shot prompting (Kojima et al., 2022) serves as a baseline condition and reflects minimally guided use of generative AI. In this condition, models are presented with the task only, typically resulting in direct answer-oriented responses with limited explicit reasoning. Chain-of-Thought prompting targets the procedural dimension of problem solving by explicitly requesting stepwise reasoning (e.g., “Let’s go step by step to make sure we have the right answer”). This technique aims to externalize intermediate reasoning processes and aligns with problem-solving frameworks that emphasize tracing decisions, validating intermediate results, and making reasoning processes visible (Wang et al., 2022; Wei et al., 2023). Persona prompting shifts the focus from reasoning structure to pedagogical framing, by assigning the model an instructional role, such as that of a mathematics teacher. Typical Persona instructions ask the model to highlight key concepts, address common student errors, discuss alternative approaches, and use learner-appropriate language. While prior findings on performance gains are mixed (Olea et al., 2024; Zheng et al., 2024), Persona prompting is expected to activate pedagogical-contextual features of solution quality, motivating its central role in research question 3 (RQ3, see 2.6). Retrieval-Augmented Generation (RAG) represents a different intervention logic by grounding model outputs in external, domain-specific knowledge (Baek et al., 2023; Gao et al., 2023; Lewis et al., 2020; Ma et al., 2023; Zhao et al., 2024). In this study, a naïve RAG approach was implemented by injecting a condensed document on problem-solving heuristics based on Pólya’s “How to Solve It” (1957) into the prompt context (Gao et al., 2023; Schorcht et al., 2025) and instructing models to explicitly consult and apply these heuristics. RAG thus targets the integration of external heuristic knowledge beyond the model’s internal parameters into solution pathways and forms the basis of RQ4 (see 2.6).

Together, the four prompt techniques operationalize complementary dimensions of instructional interventions: mini-

mal input (Zero Shot), procedural scaffolding and reasoning (Chain-of-Thought), pedagogical role activation (Persona), and knowledge grounding (RAG). Detailed prompt formulations and implementation examples are provided in Appendix 2.

2.5 Theoretical framework

This study adopts a human-centered evaluation framework emphasizing the role of expert judgment in assessing the consistency, accuracy, and educational value of AI-generated outputs, similar to frameworks and comparable studies in medicine (Tam et al., 2024). The framework is based on a theoretical and a methodological foundation, and structured around three dimensions that jointly shape the generation and evaluation of AI-generated worked-out solutions (see Fig. 1).

The conceptual foundation draws on established problem-solving theory for the study. It informs what constitutes high-quality problem solving beyond correctness, including strategic reasoning, representational use, and instructional support for learners.

Based on this conceptual foundation, the study focuses on three dimensions, specifically task characteristics, model architecture, and prompt engineering. The first dimension, task characteristics, concerns mathematical tasks, which differ systematically in their cognitive, heuristic, and representational demands, particularly across the domains of number and algebra. These task-specific features are expected to constrain both model performance and the effectiveness of prompt-based interventions, making task type a central factor in the analytical design. The second dimension concerns model architecture of LLMs. A central distinction is made between standard and reasoning models (Wei et al., 2023). These architectural differences are expected to influence both the reliability of solutions and the explicitness of reasoning processes. To examine such effects

systematically, the study contrasts two standard and two reasoning models under identical task and prompt conditions. The third dimension comprises prompt techniques, conceptualized as instructional interventions that shape how model capacities are expressed in solution texts. While model architecture constrains the general reasoning potential of an LLM, prompt design influences the structure, transparency, and pedagogical framing of generated solutions. These dimensions are analytically linked through a fully crossed design in which tasks, models, and prompt techniques are systematically varied.

The methodological foundation, human-centered quality evaluation, provides the basis for comparing outputs across systematically varied combinations of tasks, models, and prompts. Content-related, process-related, and pedagogical-contextual criteria operationalize mathematics educational quality.

2.6 Research questions

This study investigates how the mathematics educational quality of AI-generated worked-out solutions is shaped by the interaction of task characteristics, model architecture and prompt engineering techniques. The following research questions guide the analysis:

RQ1: How do variations in LLM type, prompt-engineering technique, and task type affect the *content-related quality* of AI-generated worked-out mathematical solutions?

RQ2: How do variations in LLM type, prompt-engineering technique, and task type affect the *process-related quality* of AI-generated worked-out mathematical solutions?

RQ3: To what extent does *Persona prompting* enhance the *pedagogical-contextual quality* of AI-generated worked-out solutions?

RQ4: How does *Retrieval-Augmented Generation (RAG)* influence the integration of domain-specific and contextual knowledge in AI-generated worked-out solutions?

Methodologically, RQ1 and 2 will be addressed through binary logistic regression analyses, while RQ3 and 4 will be addressed through targeted regression analyses focusing on pedagogical-contextual criteria.

3 Method

Our study follows a simulation-based design, understood here as a methodological approach in which artificial systems are systematically exposed to controlled and repeated input conditions in order to analyze the output quality (Tam et al., 2024; Laskar et al., 2023). Following our theoretical framework, we applied a fully crossed sampling grid meaning that every combination of model type, task, and

prompt technique was included in the design. Specifically, four LLMs – two standard (Claude 3.5 Sonnet, Gemini 1.5 Pro [Advanced]) and two reasoning models (ChatGPT-o3 mini, DeepSeek-R1) – were combined with six mathematical problem-solving tasks (three number and three algebra) and four prompt techniques (Zero Shot, Chain-of-Thought, Persona, and RAG). Each of the resulting 96 cells ($4 \times 6 \times 4$) was replicated 30 times, yielding a total of 2880 model outputs.

3.1 Data collection

Prior to each input, the LLM chat interface was reset, thereby eliminating potential carry-over effects from previous interactions, clearing contextual memory and ensuring consistent baseline conditions. Data collection occurred from September 2024 to May 2025 with model-specific data collection windows. The staggered data collection timeline reflects the different release and availability dates of the respective models, with the reasoning models being evaluated later in the study period. During our collection period, no major version updates of the included models were documented. Nevertheless, we acknowledge that version changes may affect performance and thus represent a potential source of bias. The theoretical considerations (see Sect. 2.4) led to the following prompt techniques employed in the present study. Zero Shot prompting served as the control condition. For the Chain-of-Thought prompting, we employed the variant from Schorcht et al. (2024) to maintain comparability. For Persona prompting, it was necessary to create a mathematics education-specific role described by Olea et al. (2024). RAG prompting primarily utilized a context injection (Schorcht et al., 2025), which can be employed by teachers without specialized programming expertise. The LLM draws upon this resource when generating problem solutions, thereby grounding its responses in relevant heuristic strategies. All prompts implemented in the study are described in Appendix 2.

3.2 Data analysis

The data analysis followed a sequential analytic design combining human expert coding with quantitative statistical analysis. In a first step, all 2880 AI-generated outputs were evaluated by trained human raters using predefined quality criteria across three dimensions: content-related, process-related, and pedagogical-contextual quality. In a second step, the resulting categorical data were subjected to quantitative analyses to examine systematic effects of model type, prompt technique, and task type.

3.2.1 Human expert rating

Human-centered evaluation followed a deductive coding scheme adapted from Küchemann et al. (2023) and Schorcht

et al. (2024). The criteria for content-related quality (*specificity, clarity, correctness*) and process-related quality (*strategy, representation, reflection*) were adopted without modification in the current study to ensure comparability of results. The pedagogical-contextual quality criteria (*call for action, situation, common mistakes, alternate approaches, citation*) were newly developed for this study and intended to capture the expected effects in the pedagogical-contextual domain that specific prompt techniques might have, especially the Persona prompt. All criteria were coded dichotomously (0 = absent; 1 = present) following the predefined coding manual from Schorcht et al. (2024) and the wide adoption of binary scales in evaluation frameworks observed by Tam et al. (2024). Precise definitions and anchor examples for these criteria can be found in Appendix 3. Partial fulfillment of a criterion was coded as 0 to ensure strict comparability. To ensure the quality and consistency of the human expert ratings, several methodological precautions were implemented. Two trained human raters independently coded a subset of responses to establish reliability benchmarks. Both raters were research assistants with a background in mathematics education. For each cell of the fully crossed sampling grid, each rater coded 60% in all variables, 20% of the material was consensually double-coded to assess reliability. Coding was refined iteratively during the period of consensual coding. The intercoder reliability varied between 0.65 and 0.99 (Cohen's κ ; see Appendix 4).

3.2.2 Statistical analysis

All analyses were conducted in Jamovi 2.6.26.0 and in SPSS 29. To identify differences across conditions, we first analyzed frequency distributions (see Sect. 4.1 for RQ1 and RQ2). We therefore aggregated the dichotomous codes for individual content- and process-related quality criteria into sum scores (0–3). Group differences in quality distributions were examined using chi-square tests of independence. However, due to sometimes highly skewed distributions and structural zero counts, chi-square tests were in these cases only used as exploratory indicators. Frequency inspections revealed that content scores of 3 were overwhelmingly prevalent, while process scores, though more evenly spread, still exhibited positive skew. This distribution meant that distinctions between 0, 1, and 2 added little analytical value, while the key contrast was whether a solution fully met the criterion (3) or not. To address this and ensure sufficient cell counts in the subsequent statistical analyses (Sects. 4.2 and 4.3), we dichotomized each sum score based on these frequency patterns: for both content-related quality and process-related quality, scores of 3 were coded as “high” (1), and 0–2 as “low” (0). While dichotomization is generally associated with a loss of information,

methodological discussions acknowledge that it can be justified in cases of strongly unbalanced distributions (MacCallum et al., 2002; DeCoster et al., 2009). Pedagogical-contextual criteria were similarly initially aggregated into a 0–5 sum score for a descriptive analysis (see Sect. 4.1). In the following statistical analyses with regard to RQ3 and 4, however, pedagogical-contextual quality criteria were selectively combined dichotomously depending on the prompt technique under observation (see Sect. 4.4). Finally, binary logistic regression models were fitted, to predict high vs. low quality outcomes from LLM type, prompt technique, and task type.

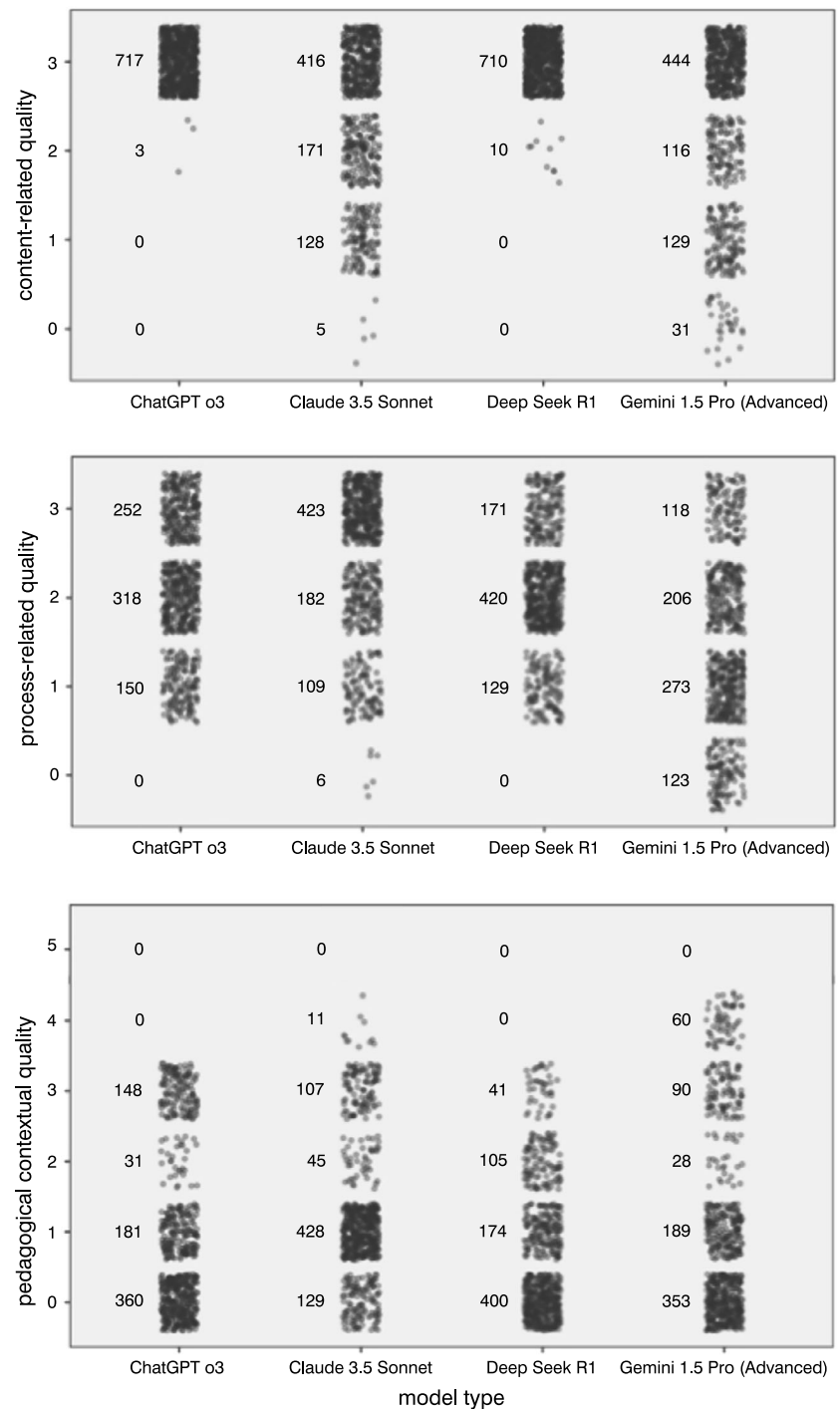
4 Results

4.1 Frequency analysis

Figure 2 illustrates the distribution of LLM performance across the three quality dimensions. The horizontal jitter of data points represents random placement for visualization, while vertical clustering indicates the actual frequency within each quality category. Chi-square tests of independence indicated significant differences in the distribution of content-related quality across models ($\chi^2(9) = 743$, $p < .001$). Reasoning models exhibited a strong concentration at the highest quality level, whereas the standard models showed more heterogeneous distributions with greater spread across intermediate levels. Notably, the reasoning models produced no responses at the quality levels 0 and 1, indicating more consistent accuracy and clarity in their solutions. Process-related quality displayed more dispersed patterns across all models, with significant distributional differences ($\chi^2(9) = 789$, $p < .001$). All models produced outputs across the full range of quality levels, indicating higher variability in strategic and representational aspects of problem solving. For pedagogical-contextual quality, no model showed a score of 5, and the highly skewed distributions and structural zero counts across quality levels violated the assumptions of chi-square tests. Reasoning models showed low performance, with also zero responses at quality level 4. Standard models showed more balanced distributions across the quality dimension: Claude 3.5 Sonnet demonstrated the most concentrated performance at quality level 1 and Gemini 1.5 Pro (Advanced) showed a very distributed pattern, with significant numbers across quality levels 1 to 4.

Figure 3 compares the performance of the four different prompt techniques. For content-related quality, all four prompt techniques achieved high scores, with most responses rated at the top level 3. The distributions differed significantly across prompt conditions ($\chi^2(9) = 152$, $p < .001$). Chain-of-Thought prompting displayed a slightly broader spread across lower levels, whereas Persona showed

Fig. 2 Distribution of outputs over all LLMs ($N = 2880$; each LLM $n = 720$ outputs) rated by content-related quality, process-related quality, and pedagogical-contextual quality



a particularly strong concentration at the highest quality level. In terms of process-related quality, the distributions differed significantly across prompt techniques ($\chi^2(9) = 323$, $p < .001$). Persona prompting showed a high proportion of high-level responses, while RAG likewise showed high frequencies at the upper levels. Zero Shot and Chain-of-Thought displayed more dispersed patterns with higher frequencies at quality levels 1 to 2. Pedagogical-contextual quality showed strongly skewed distributions across prompt

techniques, and due to very low expected frequencies in several cells, chi-square tests of independence were not defined (see Fig. 3). Zero Shot, Chain-of-Thought, and RAG exhibited extreme concentrations at lower quality levels, whereas Persona prompting displayed a broader distribution across intermediate and higher quality levels. In contrast to the other techniques, RAG showed a pronounced unimodal distribution, with the majority of responses concentrated at quality level 1.

Fig. 3 Distribution of outputs over all prompt techniques ($N = 2880$; each prompt technique $n = 720$ outputs) rated by content-related quality, process-related quality, and pedagogical-contextual quality

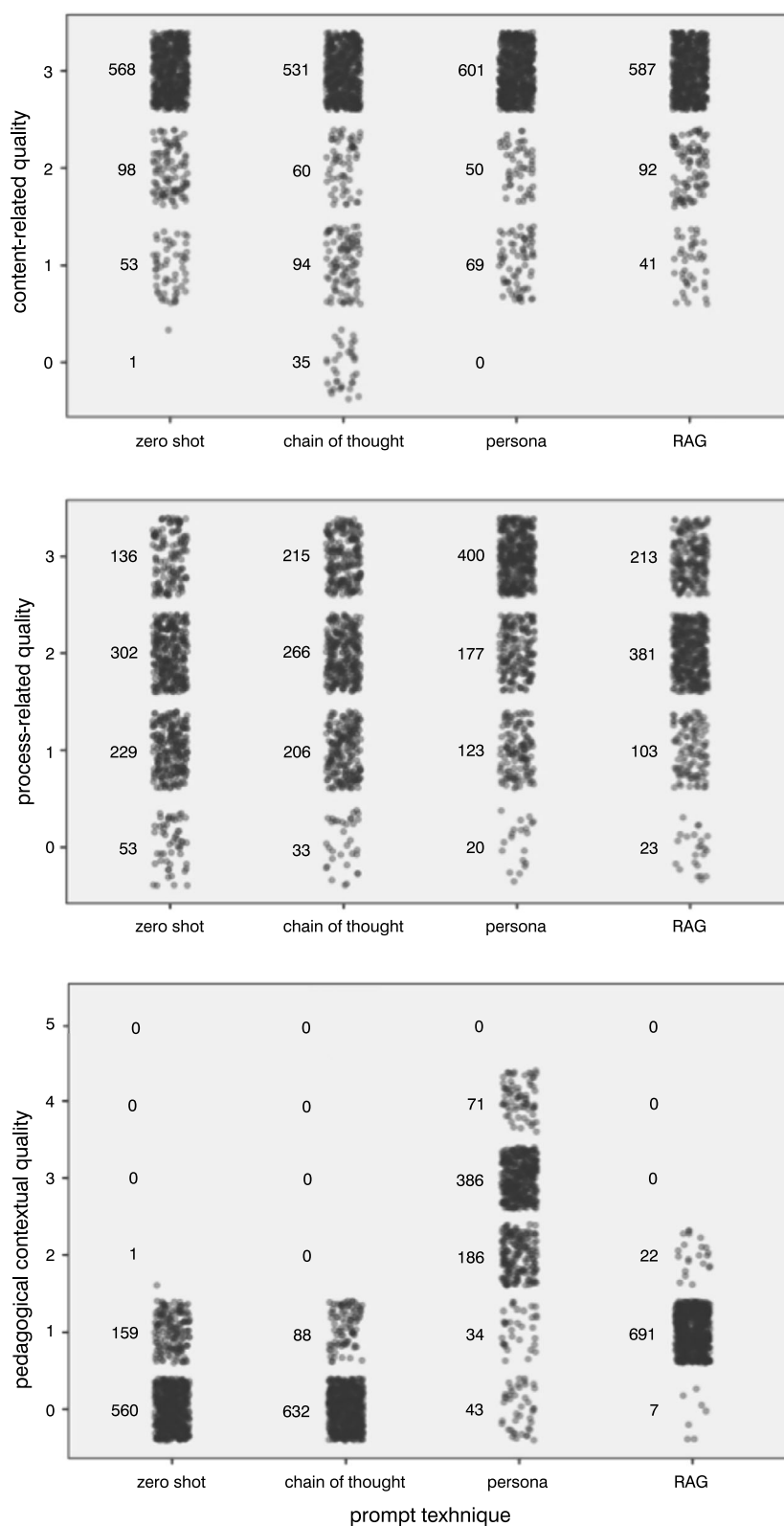


Table 1 Parameter estimates, Wald statistics, odds ratios (Exp(B)), and 95% confidence intervals for content-related quality by task type, LLMs, and prompt techniques

	B	SE	Wald	df	<i>p</i>	Exp(B)	95% CI Exp(B)	
							min.	max.
Task type (Algebra)	3.073	.151	413.007	1	<.001	21.610	16.067	29.065
ChatGPT-o3 mini			279.059	3	<.001			
Claude 3.5 Sonnet	−5.946	.593	101.006	1	<.001	.003	.001	.008
DeepSeek-R1	−1.227	.663	3.421	1	.064	.293	.080	1.076
Gemini 1.5 Pro (Advanced)	−5.684	.591	92.567	1	<.001	.003	.001	.011
Zero Shot			45.479	3	<.001			
Chain of Thought	−.649	.189	11.757	1	<.001	.523	.361	.757
Persona	.587	.190	9.558	1	.002	1.799	1.240	2.610
RAG	.337	.189	3.188	1	.074	1.401	.968	2.028
model intercept	4.868	.593	67.479	1	<.001	130.043		

4.2 Binary logistic regression analyses of content- and process-related quality

To examine predictors of content-related quality (0 = low, 1 = high), a binary logistic regression was conducted with task type, LLM type, and prompt technique as predictors (see Table 1). The full model significantly outperformed the null model, $\chi^2(7) = 1470.82$, $p < .001$; Cox & Snell R^2 was .400, and Nagelkerke R^2 was .626, indicating a substantial improvement in model fit compared to the null model. Although pseudo- R^2 measures in logistic regression do not represent explained variance in the conventional sense (Peng et al., 2002), values suggested a strong relative model performance. While the Hosmer–Lemeshow goodness-of-fit test indicated some lack of fit, $\chi^2(8) = 31.12$, $p < .001$, signaling deviations between observed and expected cell frequencies and suggesting that the model does not capture all relevant variance, this result must be interpreted cautiously given the large sample size ($N = 2880$) and the well-known sensitivity of this test to minor distributional deviations. Nevertheless, classification accuracy was high (overall 89.2%).

As Table 1 shows, algebra tasks were associated with a significant increase in the odds of high content-related quality ($OR = 21.61$) relative to number tasks, underscoring a domain effect. Relative to the reference model, ChatGPT-o3 mini, the standard models Claude 3.5 Sonnet and Gemini 1.5 Pro (Advanced) demonstrated markedly lower odds, whereas DeepSeek-R1 trended lower but was not statistically different. Concerning prompt techniques, Chain-of-Thought reduced the odds significantly and only Persona significantly outperformed Zero Shot prompts, increasing odds by 80%. RAG showed a positive but non-significant trend ($p > .05$). In summary, content-related quality was primarily driven by task domain and model choice, with prompt variants exerting more nuanced effects.

A second binary logistic regression analysis examined predictors of process-related quality (see Table 2). The full model again showed a significant improvement over the null model, $\chi^2(7) = 612.09$, $p < .001$; Cox & Snell R^2 was .191 and Nagelkerke R^2 was .266, reflecting a moderate level of explanatory power for a human-rated educational outcome. The Hosmer–Lemeshow test was again significant, $\chi^2(8) = 56.42$, $p < .001$, suggesting discrepancies between observed and expected frequencies, likely influenced by skewed distributions across predictor combinations. Classification accuracy was satisfactory, with 72.2% correctly classified cases, indicating that despite the misfit, the model captured substantial structure in the data while leaving some residual variance unexplained.

As Table 2 shows, in contrast to content-related quality, algebra tasks were associated with lower odds of high process-related quality compared to number tasks ($OR = 0.71$). Regarding LLMs, Claude 3.5 Sonnet showed nearly threefold higher odds of high process-related quality relative to ChatGPT-o3 mini, whereas DeepSeek-R1 and Gemini 1.5 Pro (Advanced) showed substantially reduced odds. All advanced prompt techniques significantly outperformed Zero Shot prompting: Chain of Thought and RAG nearly doubled the odds, while Persona even provided a sevenfold increase. These findings indicate that, while the task type and LLM type remain influential, employing sophisticated prompt techniques, specifically Persona, markedly enhances process-related quality.

4.3 Binary logistic regression predicting pedagogical-contextual quality under persona

To address the pronounced imbalance in the coding of certain pedagogical-contextual quality criteria, which appeared almost exclusively under Persona (see Fig. 3), we restricted the dataset to the 720 observations generated with Persona

Table 2 Parameter estimates, Wald statistics, odds ratios (Exp(B)), and 95% confidence intervals for process-related quality by task type, LLMs, and prompt techniques

	B	SE	Wald	df	p	Exp(B)	95% CI Exp(B)	
							min.	max.
Task type (Algebra)	−.336	.089	14.359	1	<.001	.714	.600	.850
ChatGPT-o3 mini			326.294	3	<.001			
Claude 3.5 Sonnet	1.094	.116	88.795	1	<.001	2.985	2.378	3.747
DeepSeek-R1	−.614	.124	24.315	1	<.001	.541	.424	.691
Gemini 1.5 Pro (Advanced)	−1.123	.135	69.515	1	<.001	.325	.250	.850
Zero Shot			245.554	3	<.001			
Chain of Thought	.690	.134	26.484	1	<.001	1.994	1.533	2.594
Persona	1.951	.133	216.447	1	<.001	7.034	5.424	9.121
RAG	.675	.134	14.359	1	<.001	1.964	1.509	2.555
model intercept	−1.343	.131	105.695	1	<.001	.261		

Table 3 Parameter estimates, Wald statistics, odds ratios (Exp(B)), and 95% confidence intervals for pedagogical-contextual quality under the prompt technique Persona by task type and LLMs

	B	SE	Wald	df	p	Exp(B)	95% CI Exp(B)	
							min.	max.
Task type (Algebra)	−.211	.180	1.365	1	.243	.810	.569	1.153
ChatGPT-o3 mini			153.115	3	<.001			
Claude 3.5 Sonnet	−.890	.250	12.618	1	<.001	.411	.251	.671
DeepSeek-R1	−2.759	.264	108.988	1	<.001	.063	.038	.106
Gemini 1.5 Pro (Advanced)	.078	.280	.078	1	.780	1.081	.625	1.870
model intercept	1.640	.218	56.841	1	<.001	5.157		

prompting. Four relevant criteria, call for action, situation, common mistakes, and alternate application were aggregated into sum scores ranging from 0 to 4. Importantly, the criterion *citation* was excluded, as it did not occur under Persona prompting. Frequency analyses revealed only sporadic nonzero codes for other prompt techniques. Including all conditions would have produced inflated odds ratios and unstable coefficient estimates due to these sparse cells. Sum scores were dichotomized into low (0–2) and high (3–4) pedagogical-contextual quality to ensure sufficient cell counts for stable estimation.

Since the prompt technique was fixed, a binary logistic regression was then conducted with two nominal predictors of high pedagogical-contextual quality, LLM and task type. The full model significantly improved model fit over the intercept-only model, $\chi^2(4) = 190.91$, $p < .001$, with moderate explanatory power (Cox & Snell $R^2 = .233$; Nagelkerke $R^2 = .319$). As in previous analyses, the Hosmer–Lemeshow test was significant, $\chi^2(6) = 56.41$, $p < .001$, indicating modest discrepancies between observed and expected cell frequencies. Despite this, the classification accuracy was satisfactory (77.1%), indicating that the model captures substantial systematic structure in the data. Given

the smaller, filtered sample and categorical nature of the predictors, some imbalance was anticipated; nonetheless, the overall fit remained acceptable.

As shown in Table 3, LLM type had a substantial effect: relative to ChatGPT-o3 mini, both Claude 3.5 Sonnet and DeepSeek-R1 showed significantly reduced odds of high pedagogical-contextual quality under Persona prompting, whereas Gemini 1.5 Pro (Advanced) did not differ significantly. The task type did not exert a significant effect, suggesting that, within Persona prompting, differences in pedagogical-contextual quality are primarily attributable to the LLM rather than the mathematical domain.

4.4 Binary logistic regression on citation under RAG

A binary logistic regression targeting the dichotomous *citation* criterion was conducted exclusively under the RAG prompting condition. However, inferential modeling was not feasible due to extreme outcome imbalance: Of the 720 cases analyzed, more than 99% were coded as containing a citation. This near-deterministic distribution in the RAG condition is visible in Fig. 3 (note that discrepancies occur due to different pedagogical-contextual quality criteria

being aggregated). Such quasi-perfect separation may produce infinite or highly unstable maximum likelihood estimates, and thereby precludes any meaningful statistical inference (see e.g., Heinze & Schemper, 2002). Accordingly, no regression results are reported for this criterion. Instead, this outcome corroborates the descriptive finding that RAG prompting almost invariably elicits explicit references to the provided heuristic document, confirming its function as a retrieval-based prompting technique rather than one relying solely on internal model reasoning.

5 Discussion

Regarding RQ1, the results partly replicate the findings by Schorcht et al. (2024) and show that content-related quality is primarily determined by the choice of LLM. Architectural differences between standard and reasoning models have impacts on content-related quality. In contrast, only Persona prompting showed a statistically significant, yet moderate effect. A closer comparison of the two task types in this study suggests that task characteristics may exert an influence on content-related quality, favoring algebra tasks. However, this observation should be interpreted cautiously, as only three tasks per domain were included. A plausible explanation is that algebra problems often follow well-defined, algorithmic solution paths, such as setting up and manipulating equations, which current LLMs have become increasingly adept at standardizing and executing correctly. While the substantial number of replications per task (480 times) strengthens internal reliability, the limited variety of tasks constrains the generalizability of our findings. Our preliminary evidence suggests that the more a problem aligns with a clear, formulaic procedure, the more reliably current (reasoning) models can reproduce it without omission or error. This supports previous studies showing that LLM performance deteriorates as problem contexts become more complex or semantically demanding (Dao & Le, 2023; Wardat et al., 2023; Spreitzer et al., 2024). However, further research with a broader range of tasks would be needed to confirm whether this pattern holds consistently across these mathematical domains. Within our framework, this aligns the model dimension with content-related quality: architectural differences (standard vs. reasoning) primarily determine mathematical accuracy, whereas prompting acts as a weaker secondary lever.

In contrast, RQ2 reveals a different pattern: process-related quality is strongly influenced by prompt techniques, again, in parts confirming Schorcht et al. (2024). All three advanced prompting techniques improved process-related quality significantly relative to Zero Shot. Persona showed the strongest effect with a more than sevenfold improvement in odds, followed by Chain of Thought and RAG. Sur-

prisingly, algebra tasks reduced the odds for high process-related quality compared to number tasks. One possible explanation is that number problems more often require multi-step heuristic approaches, resulting in richer process descriptions.

Regarding RQ3, Persona prompting systematically activated pedagogical-contextual features that were largely absent under other prompting conditions. Because such features appeared almost exclusively under Persona, the regression analysis was intentionally limited to this subset ($n = 720$) to avoid distorted estimates. Within the Persona condition, the regression revealed only limited model differences not clearly favoring standard or reasoning models. Relative to ChatGPT-o3 mini, Claude 3.5 Sonnet and DeepSeek-R1 showed significantly lower odds of achieving high pedagogical-contextual quality, while Gemini 1.5 Pro (Advanced) did not differ significantly from the reference model. No significant effects were found for task type, indicating that pedagogical-contextual quality does not systematically vary between number and algebra problems once the pedagogical role is activated. The results imply that Persona functions as a general pedagogical scaffold rather than interacting strongly with mathematical domain or model architecture. However, an important didactic caveat remains: high pedagogical quality is not necessarily aligned with mathematical correctness. Persuasive but incorrect explanations occurred, underscoring the need for content verification before classroom use.

For RQ4, RAG achieved a near-perfect success in producing explicit citations of the provided Pólya heuristics. With 99% of all RAG responses containing explicit references to the source document, RAG demonstrates seemingly reliable functionality. However, whether the cited knowledge was actually accessed or whether only a citation was hallucinated by the LLM in response to the requirement cannot be answered clearly. RAG showed significant improvements in process-related quality compared to Zero Shot. The improvement is, however, smaller than with Persona, which suggests that pure knowledge supply is less effective than the activation of specific roles. RAG showed no significant improvement in content-related quality, despite a positive trend. This is remarkable, since the provided Pólya heuristics are explicitly designed to improve mathematical problem-solving strategies. Several explanations may account for this finding: LLMs may reference the document without effectively applying its content, or their internal mathematical reasoning capabilities already limit the marginal benefit of additional heuristic information.

Several limitations must be considered when interpreting our results. First, the operationalization of mathematics educational quality through binary coding reduces complex concepts to simplified categories, potentially obscuring important nuances of pedagogical effectiveness. Al-

though binary logistic regression was used, it remains uncertain whether all relevant dependency structures (e.g., task-specific covariances) were adequately modeled. Second, the simulation-based design with 30 repetitions of identical prompts creates artificial conditions that only partially reflect classroom use, and our models showed considerable misfit, which means that the results must be interpreted with caution. Third, we base our analysis on recent LLMs, whose model versions are subject to rapid change. While our results regarding Persona and RAG may be model-specific, it cannot be ruled out that these are methodological artifacts. Nevertheless, while specific model versions may change over time, the analytical distinction between standard and reasoning-oriented LLM architectures remains stable and relevant for interpreting the present findings. Finally, the six selected problem-solving tasks represent only a narrow segment of mathematical domains, and the focus on number and algebra restricts transferability to areas such as geometry, data, or modelling (Spreitzer et al., 2024).

6 Conclusion

This study reveals that the effective use of LLMs in mathematics education requires strategic orchestration rather than reliance on a single model or prompting approach. Within the limits of our simulation-based design, model architecture emerged as the primary determinant of content-related quality, with reasoning models producing more mathematically correct and complete solutions than standard models. This suggests that, for educational contexts where correctness is critical, model selection is more consequential than modifying prompts alone. At the same time, prompt engineering proved highly effective for enhancing process-related and pedagogical-contextual quality. In particular, Persona prompting consistently activated pedagogical features such as reflection, contextualization and addressing common mistakes, indicating its potential value for teacher education and professional development and instructional design. RAG, by contrast, delivered only modest gains: While RAG can integrate up-to-date or domain-specific information, its benefit hinges on careful curation of knowledge sources. Given that high pedagogical-contextual quality can coexist with mathematical errors in our data, educational implementations may require systematic two-stage review processes: pedagogical enhancement followed by mathematical accuracy verification. Finally, task characteristics appear to set important constraints: algebra problems were handled more reliably than number tasks with respect to content-related quality, while the pattern seems reversed for process-related quality. Taken together, these findings suggest that no single approach guarantees optimal performance across all quality dimensions. Effective educational

deployment of LLMs thus requires a balanced combination of model selection, targeted prompt engineering, and careful consideration of task-specific affordances and limitations.

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Declarations

Declarations We declare that this work has not been submitted for publication elsewhere.

Ethical Declarations In accordance with local legislation and institutional requirements, an ethics board approval was not required for this study. In Germany, as stated by the German Research Association (DFG, https://www.dfg.de/foerderung/faq/geistes_sozialwissenschaft/index.html#anker13417818), the present survey study did not require the approval of an ethics committee, because the research did not have human participants resp. respondents.

Competing Interests The authors have no competing interests to declare.

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